

Handling Digital Evidence

Estimated time needed: 10 minutes

Learning Objective:

After completing this reading, you will be able to:

- Understand the importance of chain of custody in digital forensics
- Identify the steps and safeguards involved in preserving digital evidence
- Discuss the implications of breaches in the chain of custody on legal proceedings

In this reading, we will explore the concept of handling digital evidence, the chain of custody framework, and the measures needed to ensure the integrity and credibility of digital assets in a forensic investigation.

Chain of custody

Chain of custody refers to the process used to track and manage the movement of digital assets through their lifecycle. It documents every individual and organization that handles the asset, along with the date, time, and reason for each transfer. This ensures the preservation of evidence for admissibility in court.

An asset can be any digital object, such as computers, mobile devices, servers, external hard drives, or data files. Proper chain-of-custody management maintains the integrity and credibility of the evidence, ensuring it can be trusted for legal use.

However, a break in the chain of custody occurs when the control over the asset is uncertain or actions on the asset are unaccounted for. When this happens, the system or data may no longer be trusted, and its evidence may not be admissible in court.

Chain of custody framework

The NIST Chain of Custody Framework provides standards for securing and maintaining the integrity of digital evidence throughout its lifecycle.

1. Identify

- Catalog all systems, devices, software, data, and people involved.
- Develop logs that track transactions and access and assess these logs regularly for gaps.
- Inventory critical assets and external systems integrated with them.

2. Protect

- Implement both physical and electronic access controls.
- Authorize access based on job functions, applying the principle of least privilege.
- Monitor all activities and transactions continuously to protect network integrity.

3. Detect

- Track all assets using unique identifiers.
- Implement monitoring and alerting systems for detecting chain-of-custody breaches.
- Set alert thresholds and track access through electronic or physical logs.

4. Respond

- Establish a response process for breaches and forensic activities.
- Investigate detected breaches and assess whether they constitute reportable incidents.
- Ensure personnel are familiar with their roles during incident response.

5. Recover

- Develop resilience and recovery plans for restoring affected systems.
- Sanitize and review systems for modifications, ensuring validated software and firmware are used in recovery.
- Revalidate the chain of custody before reintegrating data or systems.

Tracking the chain of custody

It is essential to maintain accurate and up-to-date records for the chain of custody. Tools such as the NIST Chain of Custody Tracking Form can help ensure detailed documentation of every step in the asset's lifecycle.

Summary

In this reading, you learned that:

- Chain of custody is crucial for maintaining the integrity and admissibility of digital evidence in legal cases.
- A break in the chain of custody can lead to evidence being discredited and rejected in court.
- The NIST Chain of Custody Framework includes steps to identify, protect, detect, respond to, and recover from chain of custody breaches.
- Protecting digital evidence requires both physical and electronic safeguards, including access controls and monitoring.
- Proper documentation using tools such as the Chain of Custody Tracking Form ensures accountability at every step.

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